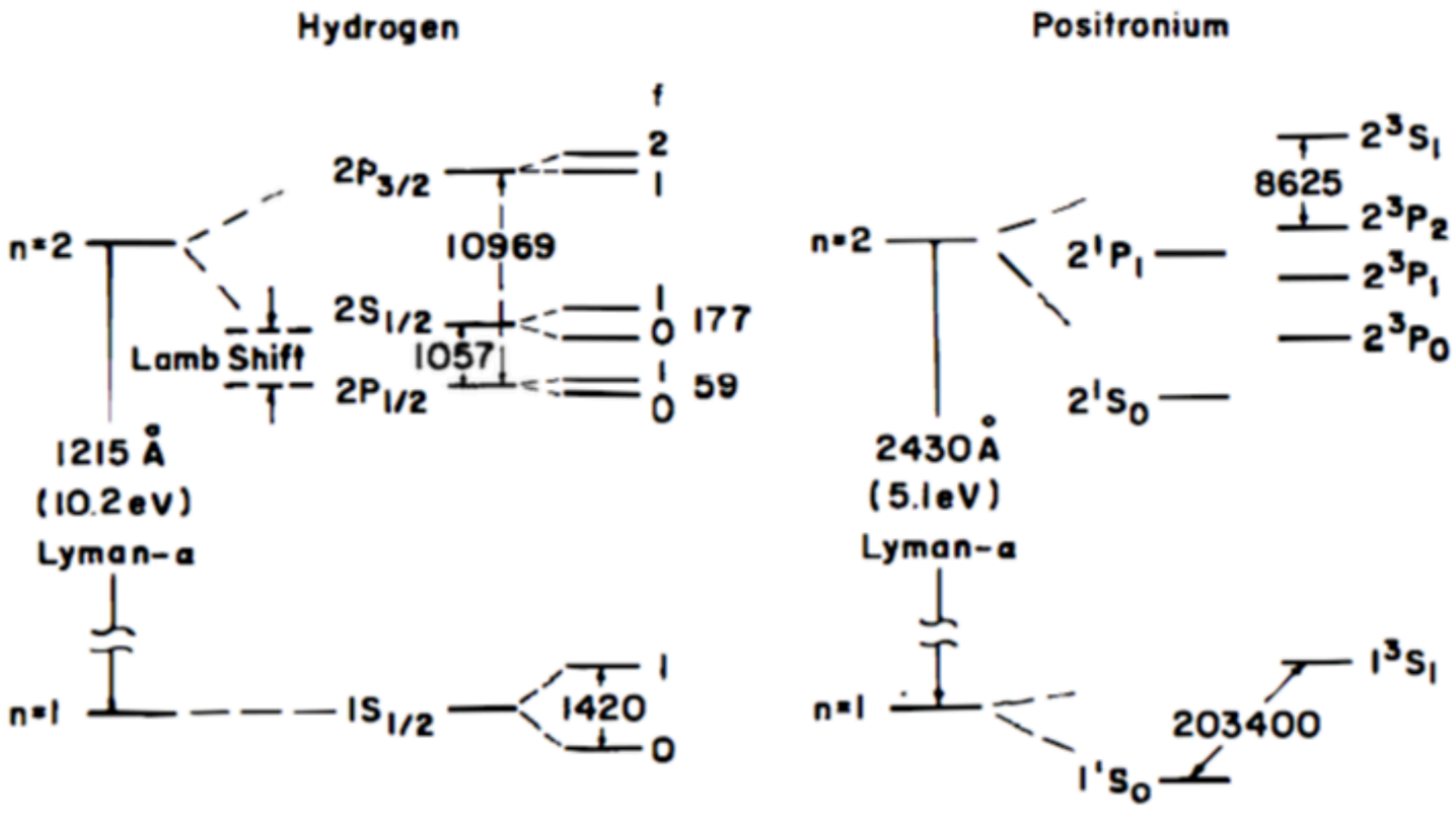


Fig. 4.1: Comparison of the 1S, 2S, and 2P energy levels of hydrogen and positronium, from (Berko and Pendleton 1980).

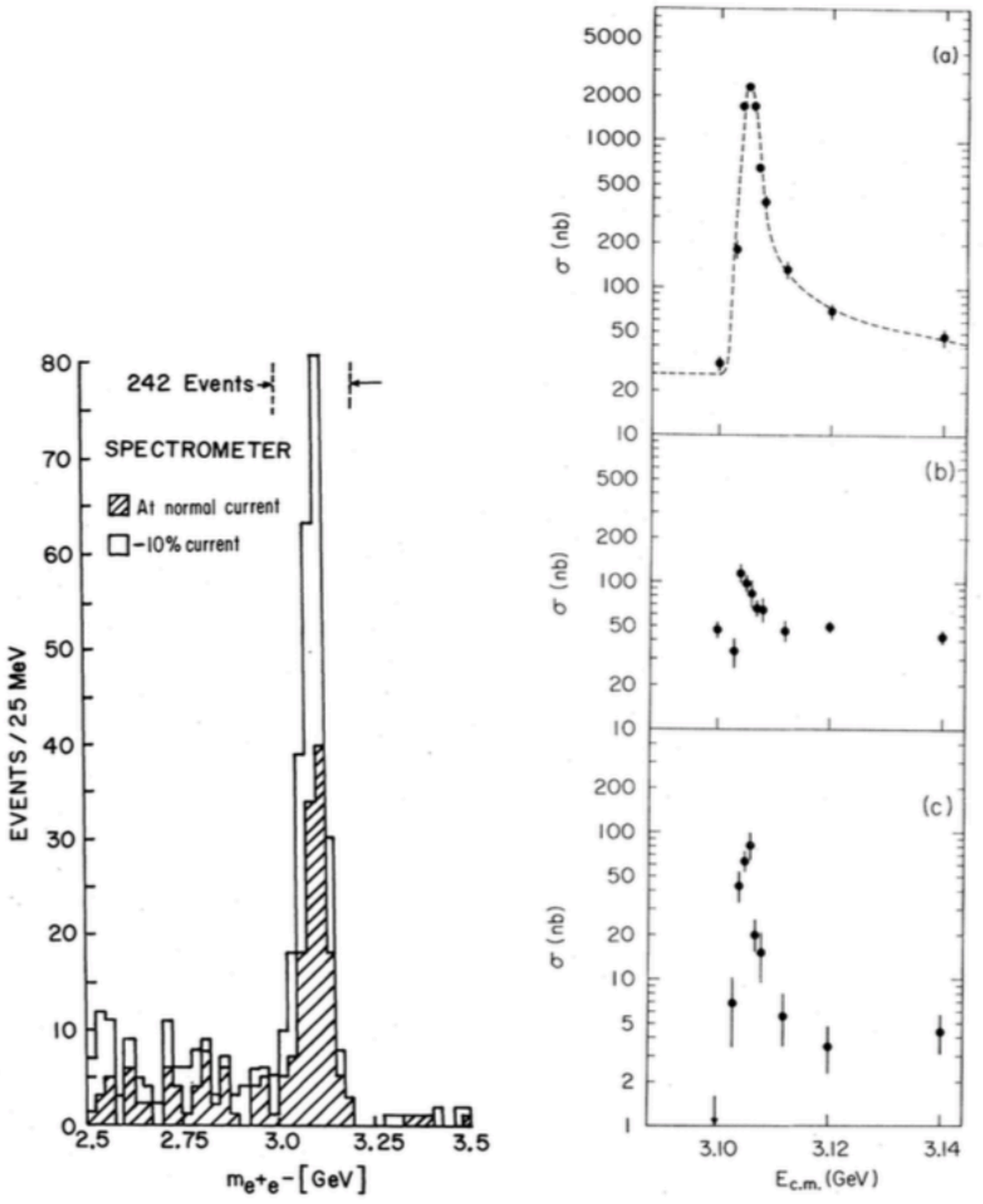


Fig. 5.4: Left: Accumulation of events from the reactions $pp \to e^+e^- + X$ at a fixed value of the e^+e^- invariant mass, proving the existence of the narrow resonance $J\psi$, from (Aubert *et al.* 1974). Right: Observation of a resonance in e^+e^- annihilation near 3.1 GeV, proving the existence of the J/ψ, from (Augustin *et al.* 1974). From top to bottom, the three plots show the production rate of final states with (a) multiple hadrons, (b) e^+e^-, and (c) pairs of μ, π, K.

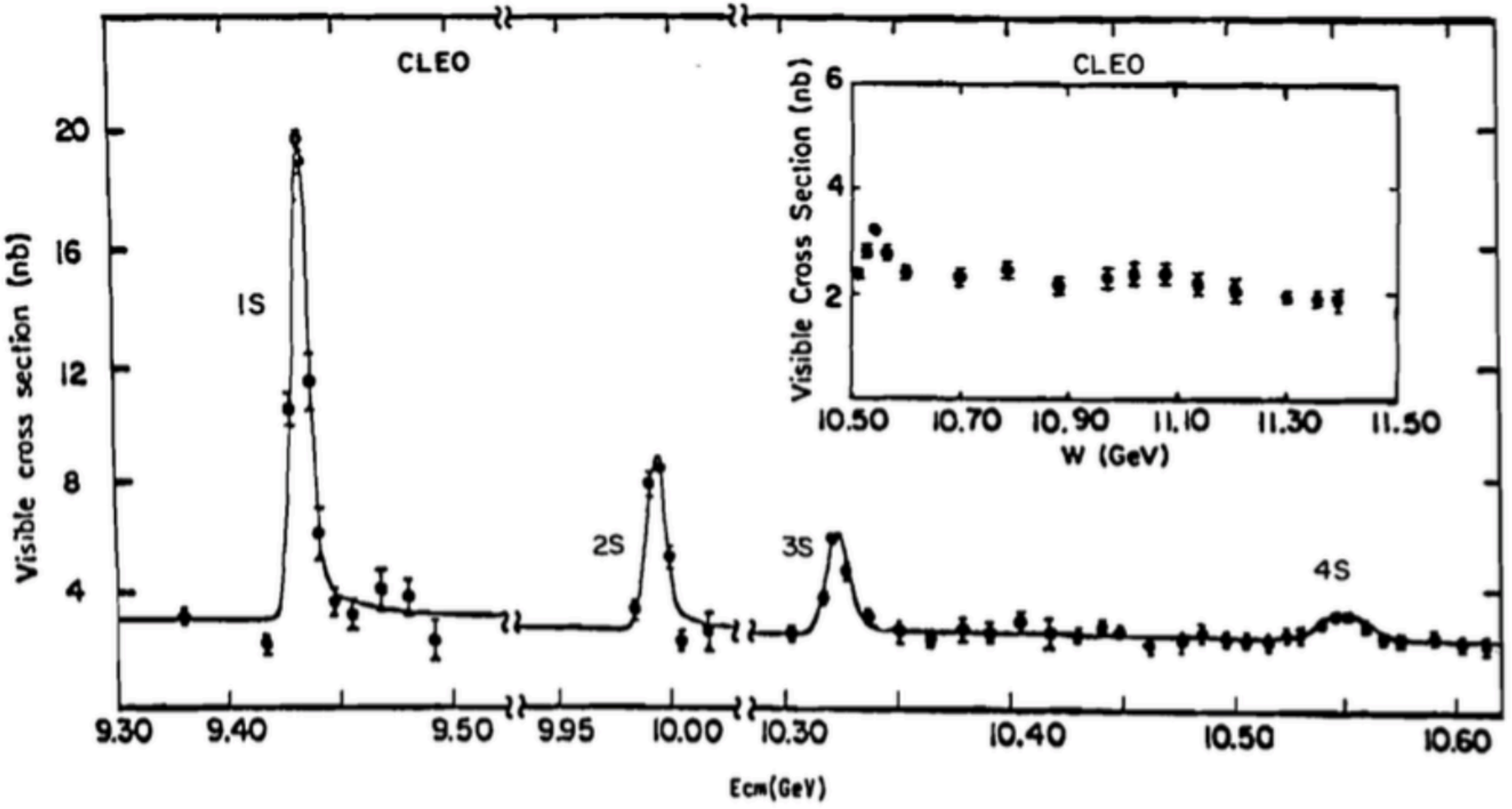


Fig. 5.5: Resonances in e^+e^- annihilation to hadrons corresponding to the S states of the Υ family, from (Silverman 1981). The inset extends the dataset to higher center of mass energy.

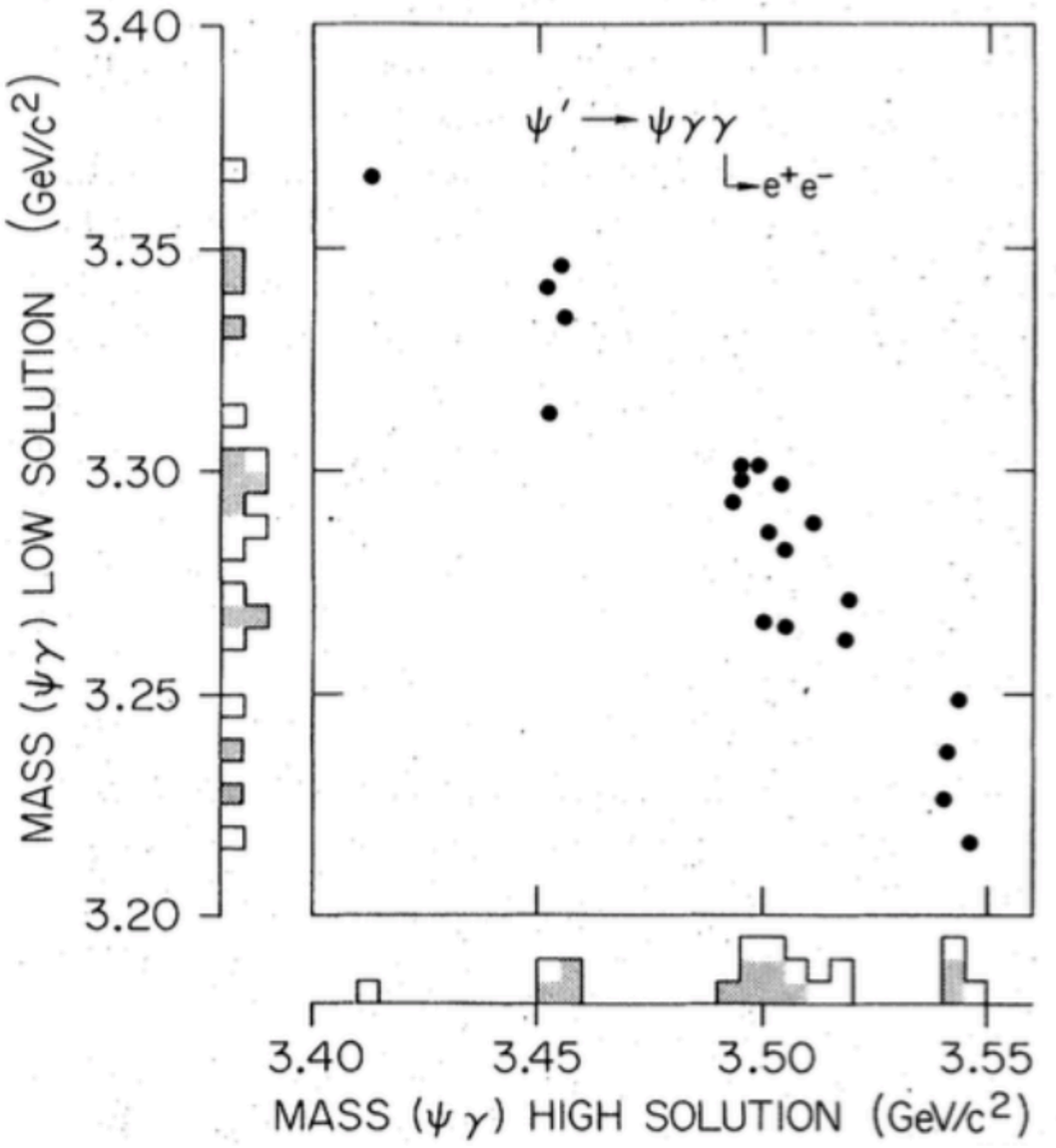


Fig. 5.6: Distribution of $\psi\gamma$ invariant masses observed in the reaction $\psi' \to \gamma\gamma\, J/\psi$ by the Mark I experiment at SPEAR, from (Tannenbaum *et al.* 1978).

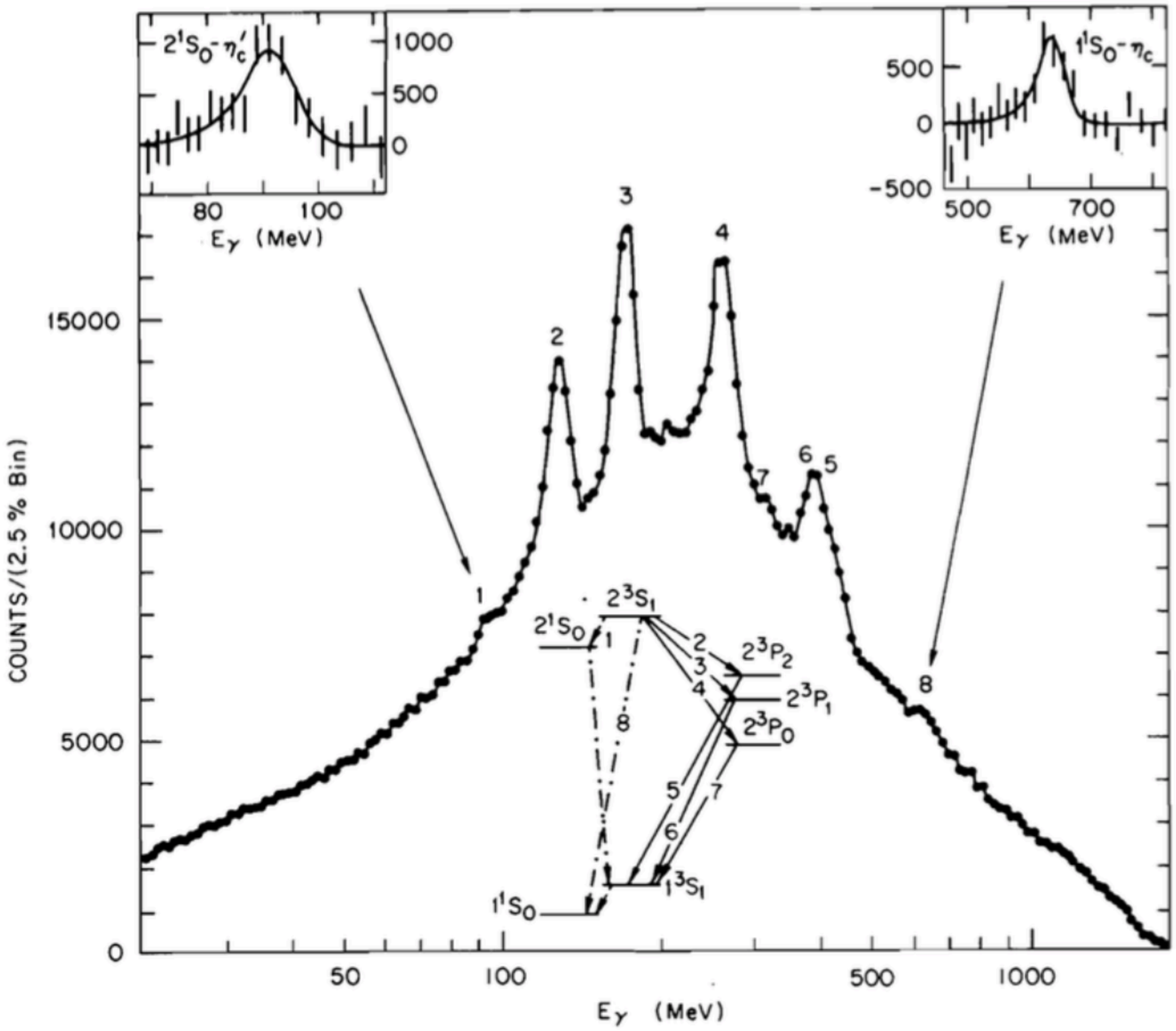


Fig. 5.7: Energy spectrum of photons observed at the ψ' resonance by the Crystal Ball experiment, from (Scharre 1981).

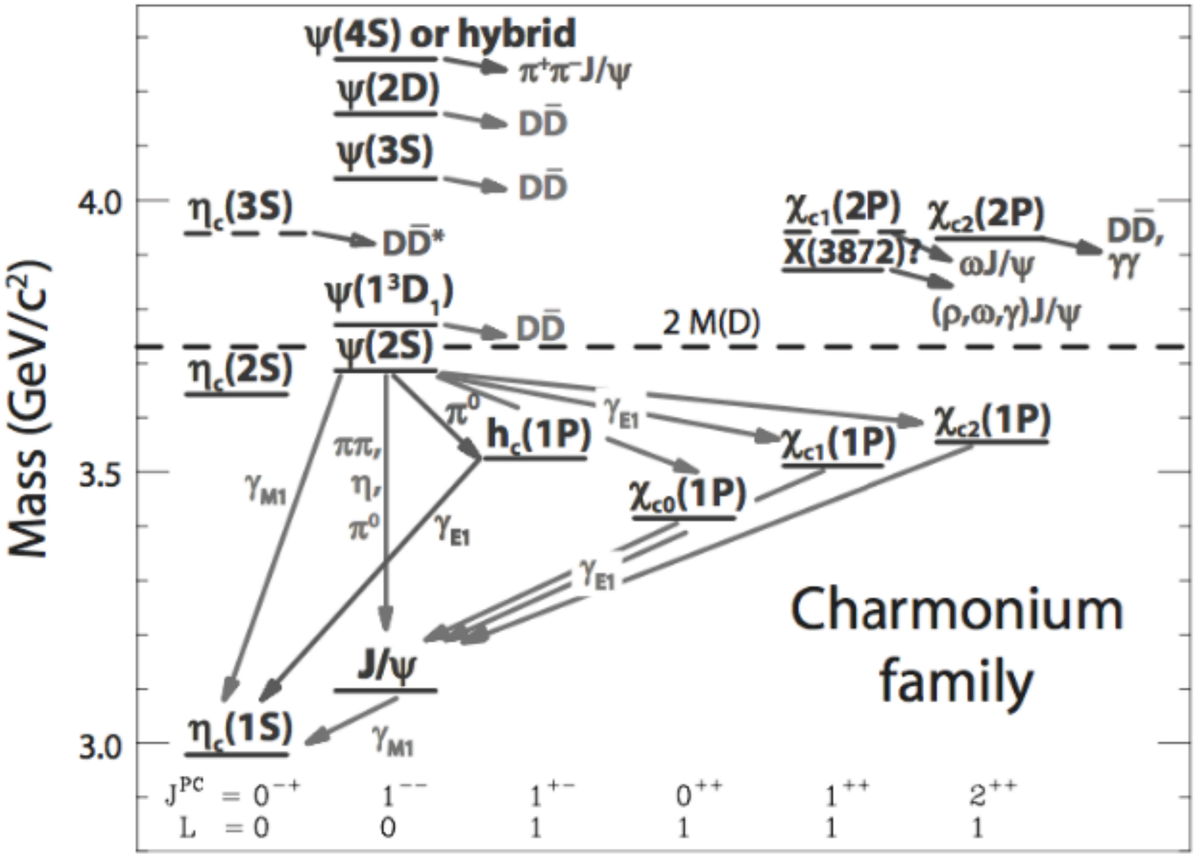


Fig. 5.8: Observed states and transitions of the J/ψ system, from (Eichten, Godfrey, Mahlke, and Rosner 2008). Note that the lowest P and D states are labelled 1P and 1D in this figure, whereas in the hydrogen atom they would be called 2P and 3D. The dashed horizontal line marks the threshold for pair production of D mesons, described in Section 5.4.

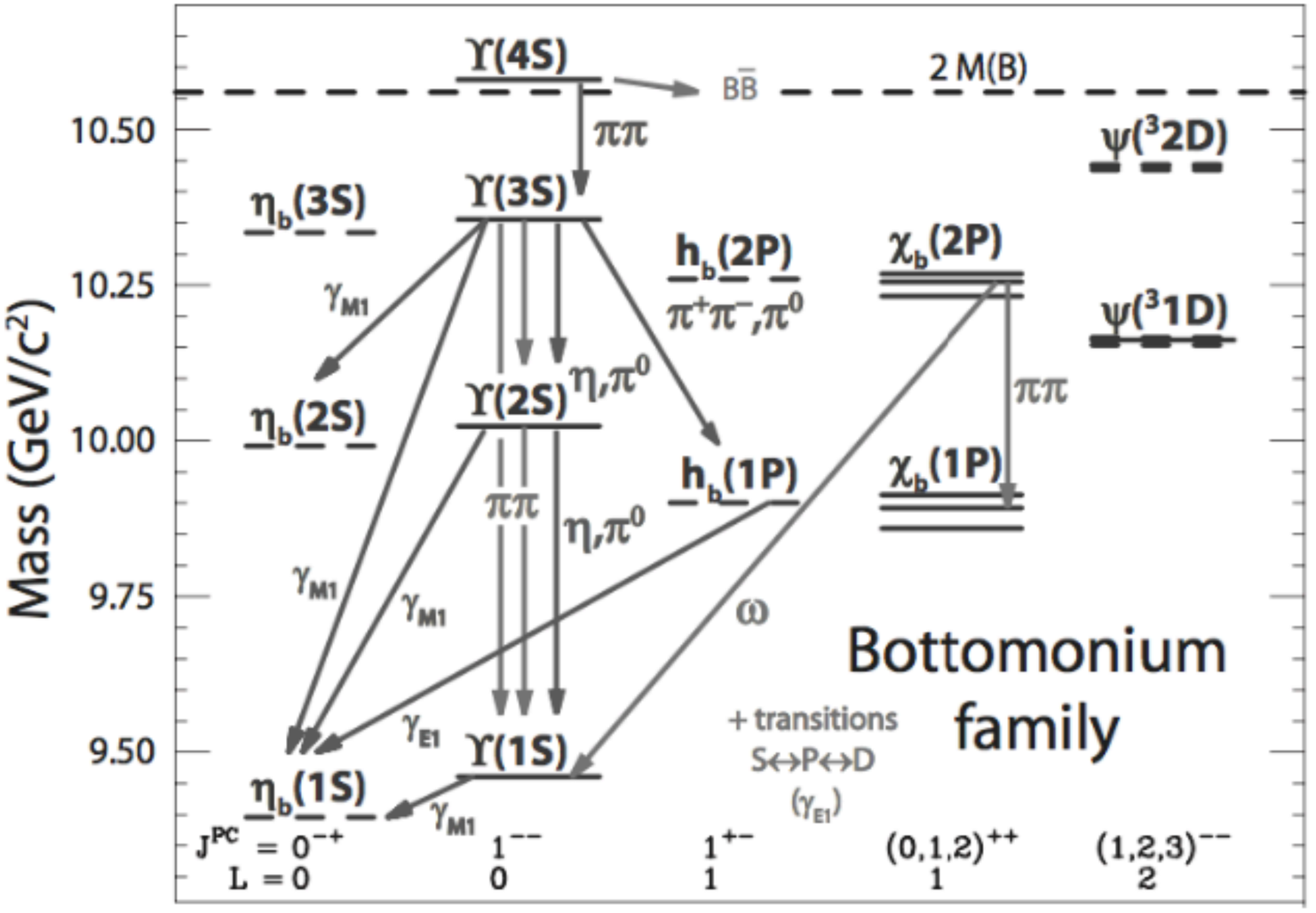


Fig. 5.9: Observed states and transitions of the Υ system, from (Eichten, Godfrey, Mahlke, and Rosner 2008). The dashed horizontal line marks the threshold for pair production of B mesons, described in Section 5.4.

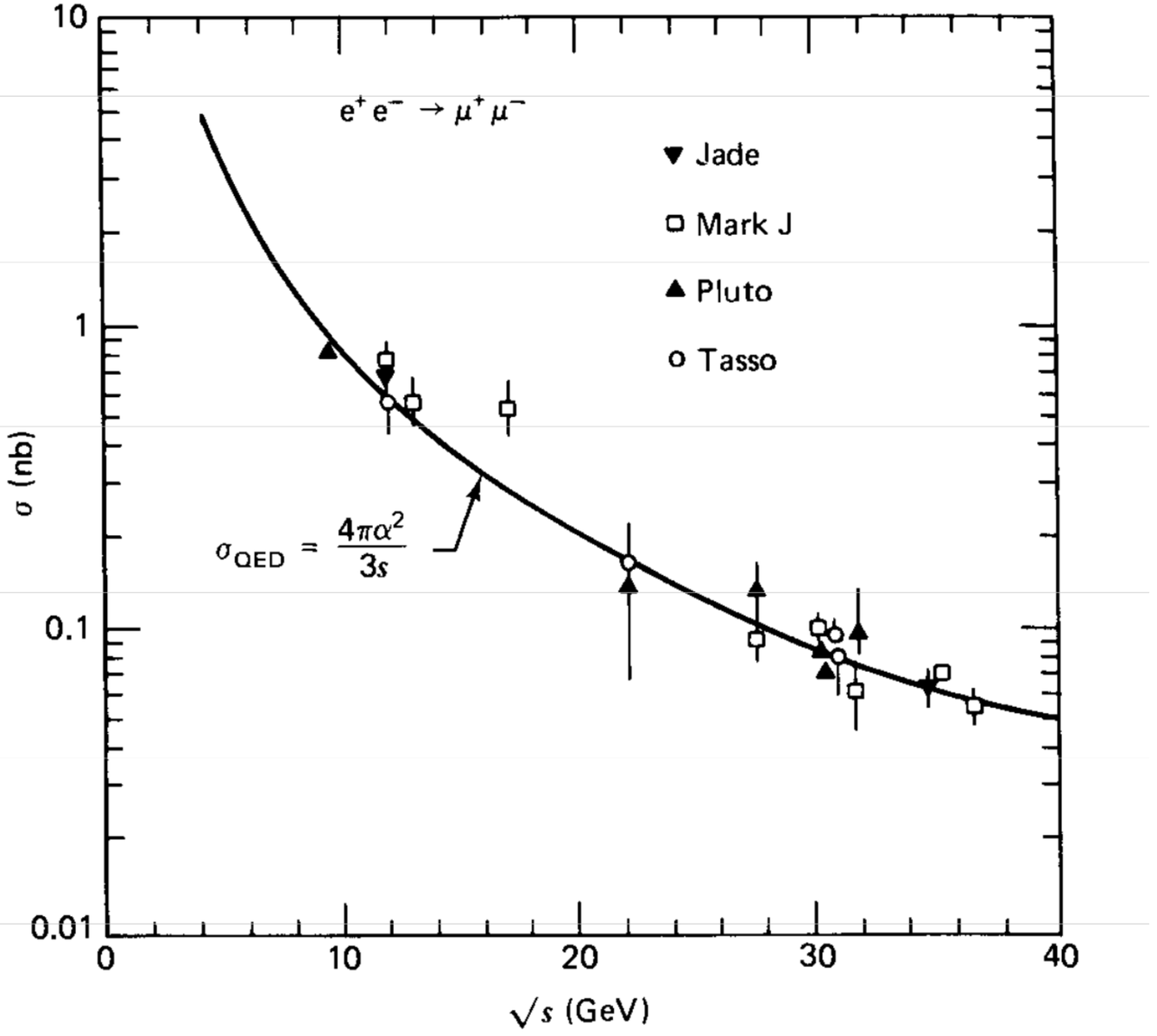


Fig. 6.6 The total cross section for $e^-e^+ \to \mu^-\mu^+$ measured at PETRA versus the center-of-mass energy.

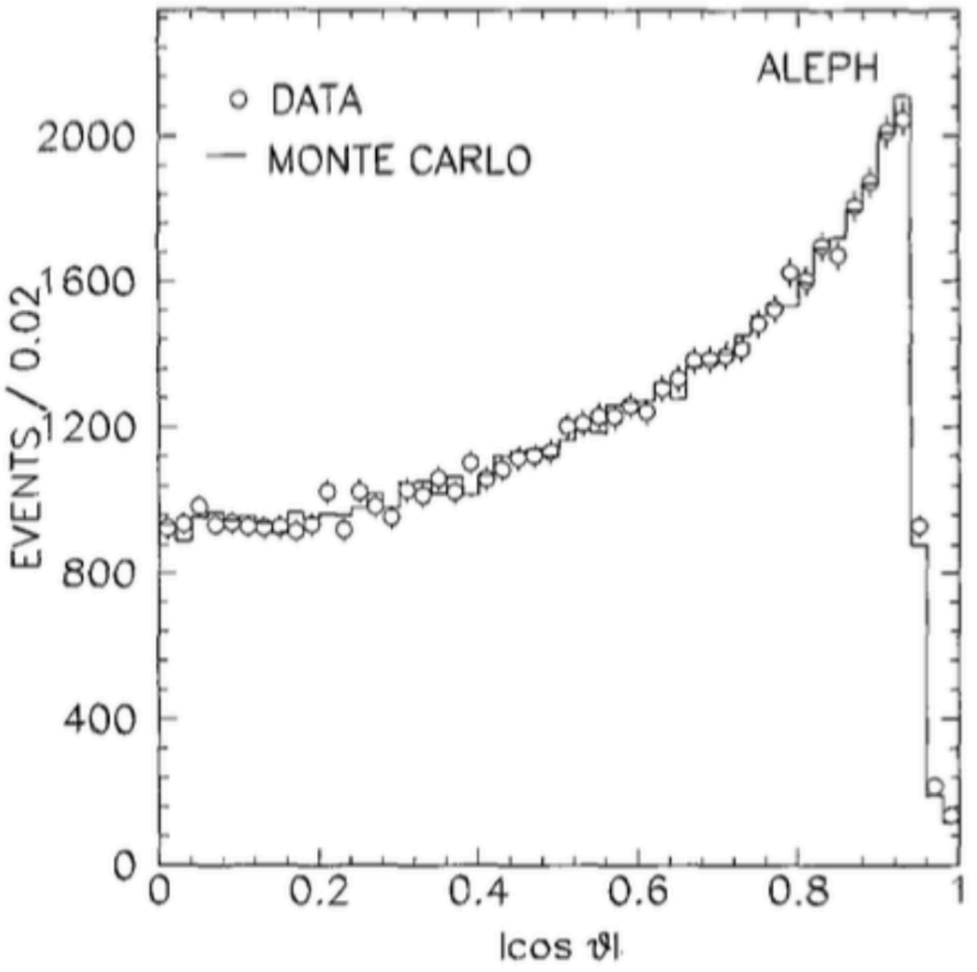


Fig. 8.3: Distribution of the orientations of jet axes in e^+e^- annihilation to hadrons as a function of $|\cos\theta|$, as measured by the ALEPH experiment at the LEP e^+e^- collider, from (Decamp *et al.* (1990)).

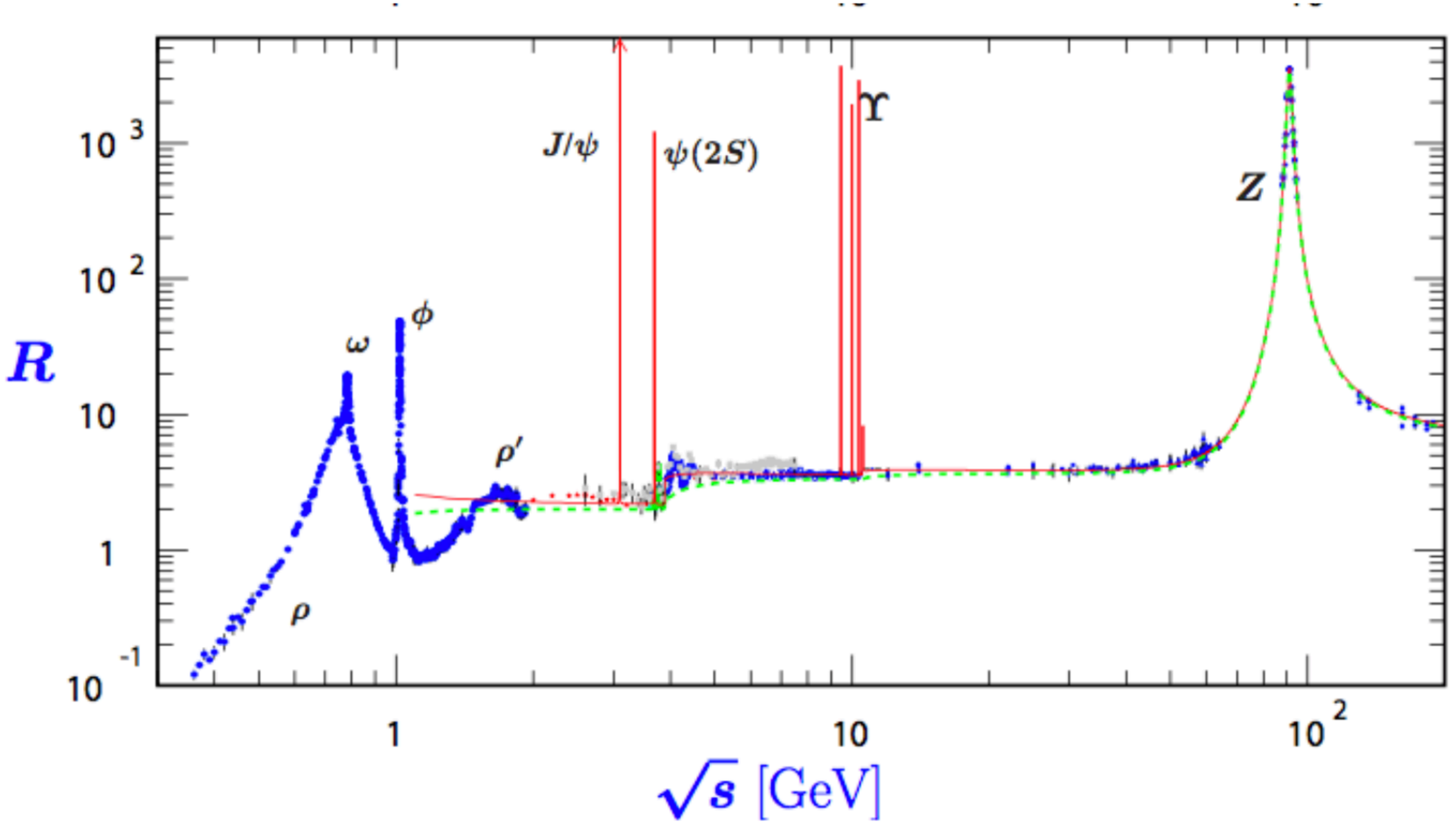


Fig. 8.1: Measurements of the total cross section for e^+e^- annihilation to hadrons as a function of energy, compiled in (Patrignani 2016). The lower figure shows the ratio $R = \sigma(e^+e^- \to \text{hadrons})/\sigma(e^+e^- \to \mu^+\mu^-)$. The green dotted curve is the prediction (8.56). The vertical red lines show the ψ and Υ resonances. The horizontal red curve is the prediction (11.72).

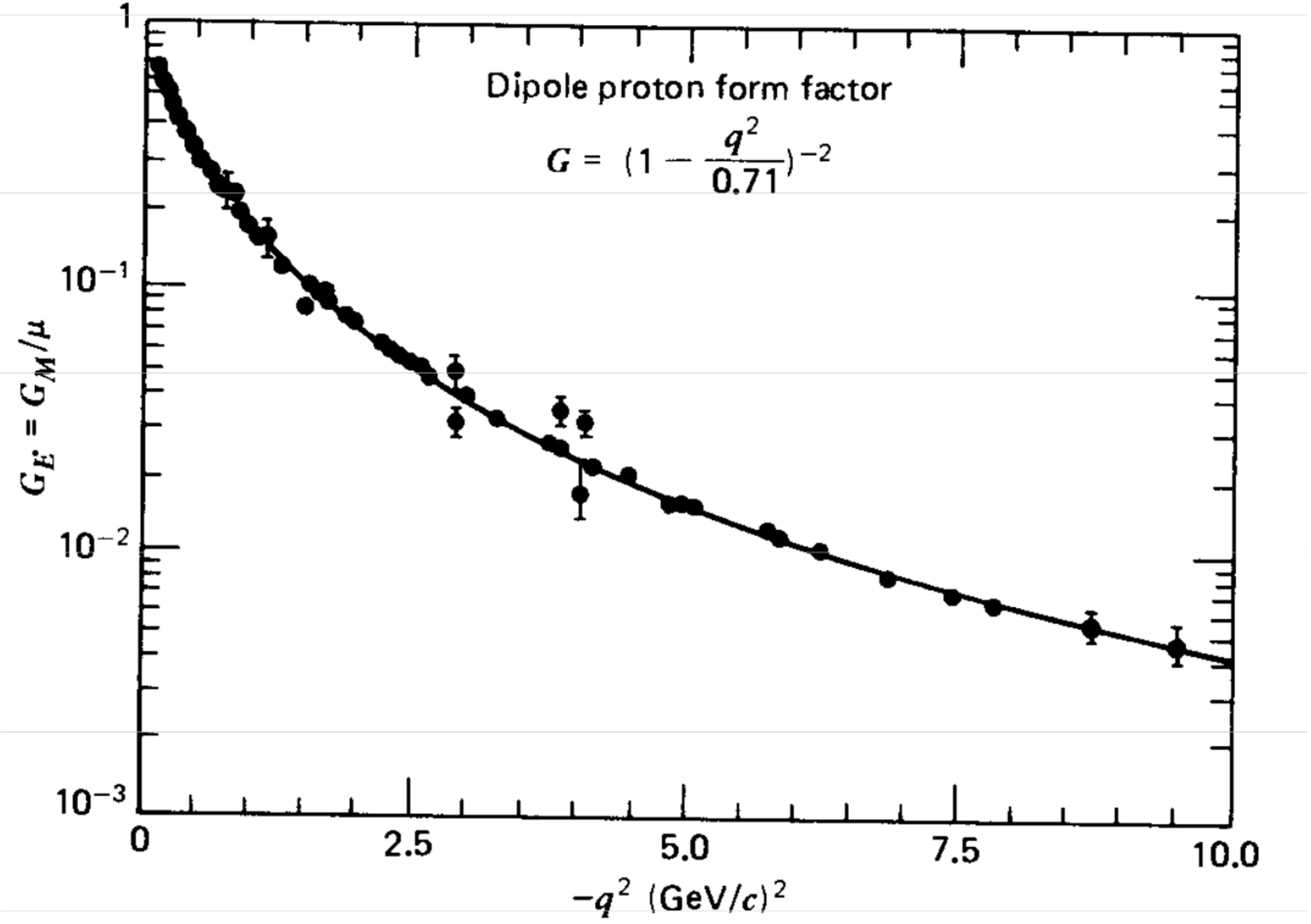


Fig. 8.4 The proton form factors as a function of q^2.

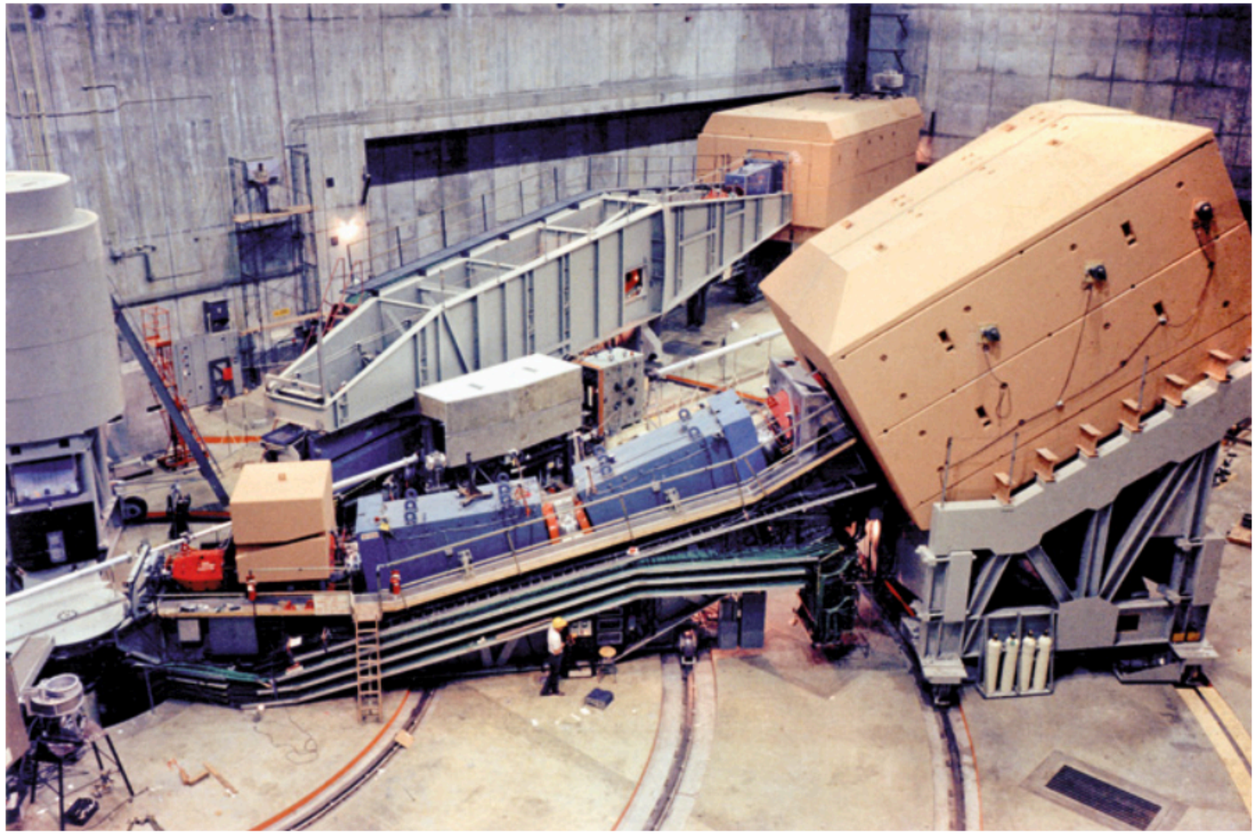


Fig. 9.1: Layout of the SLAC-MIT deep inelastic scattering experiment (figure courtesy of SLAC). Electrons strike a hydrogen target just under the cylinder on the left of the figure. Scattered electrons, moving left to right, pass through a string of magnets that measure their momenta, and then into the large electromagnetic calorimeter on the right.

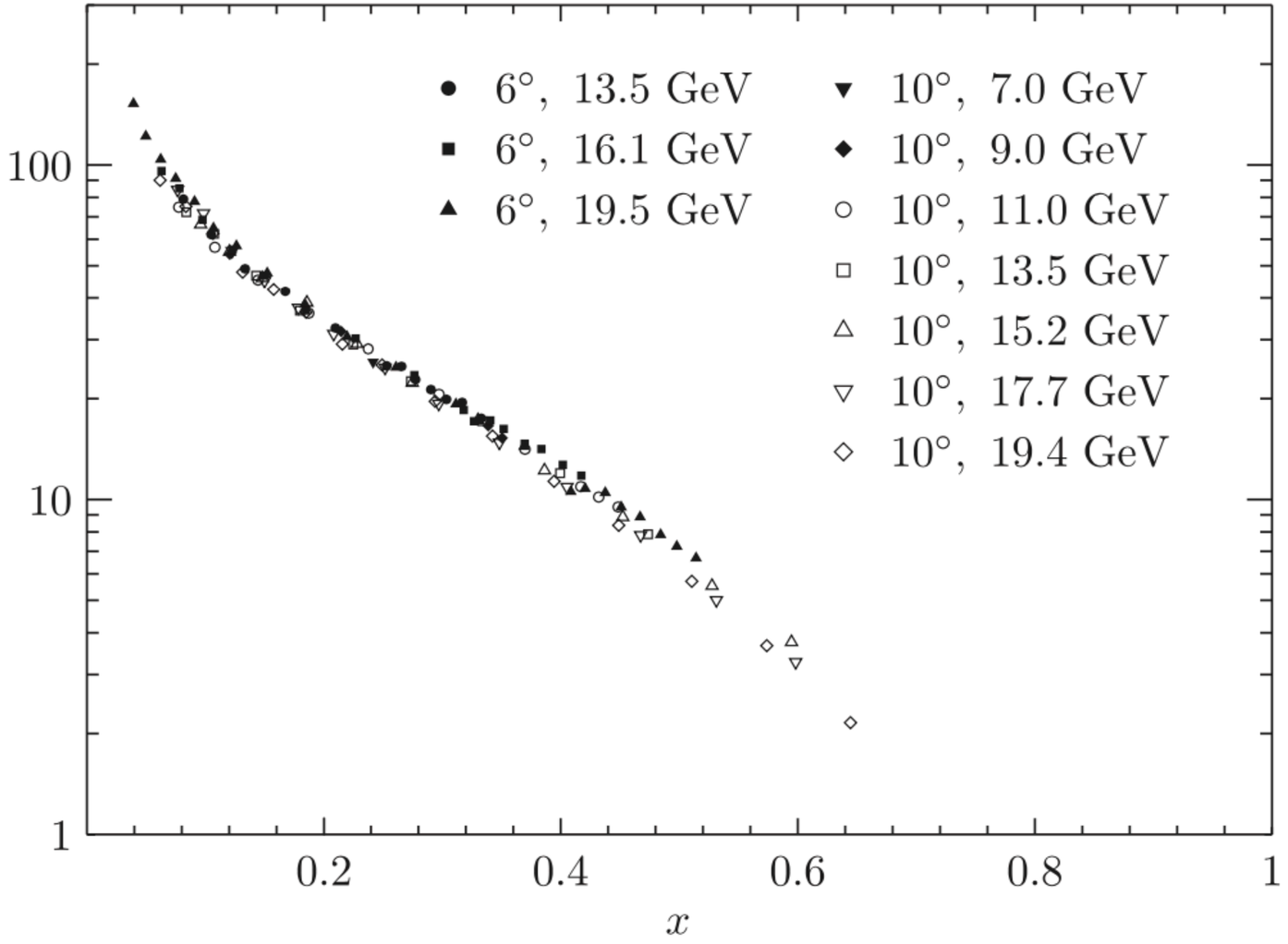


Fig. 9.4: Measurements of the quantity F_2 by the SLAC-MIT experiment, at energy and angle settings giving $Q^2 > 1$ GeV^2, plotted as a function of x, from (Peskin and Schroeder 1995).

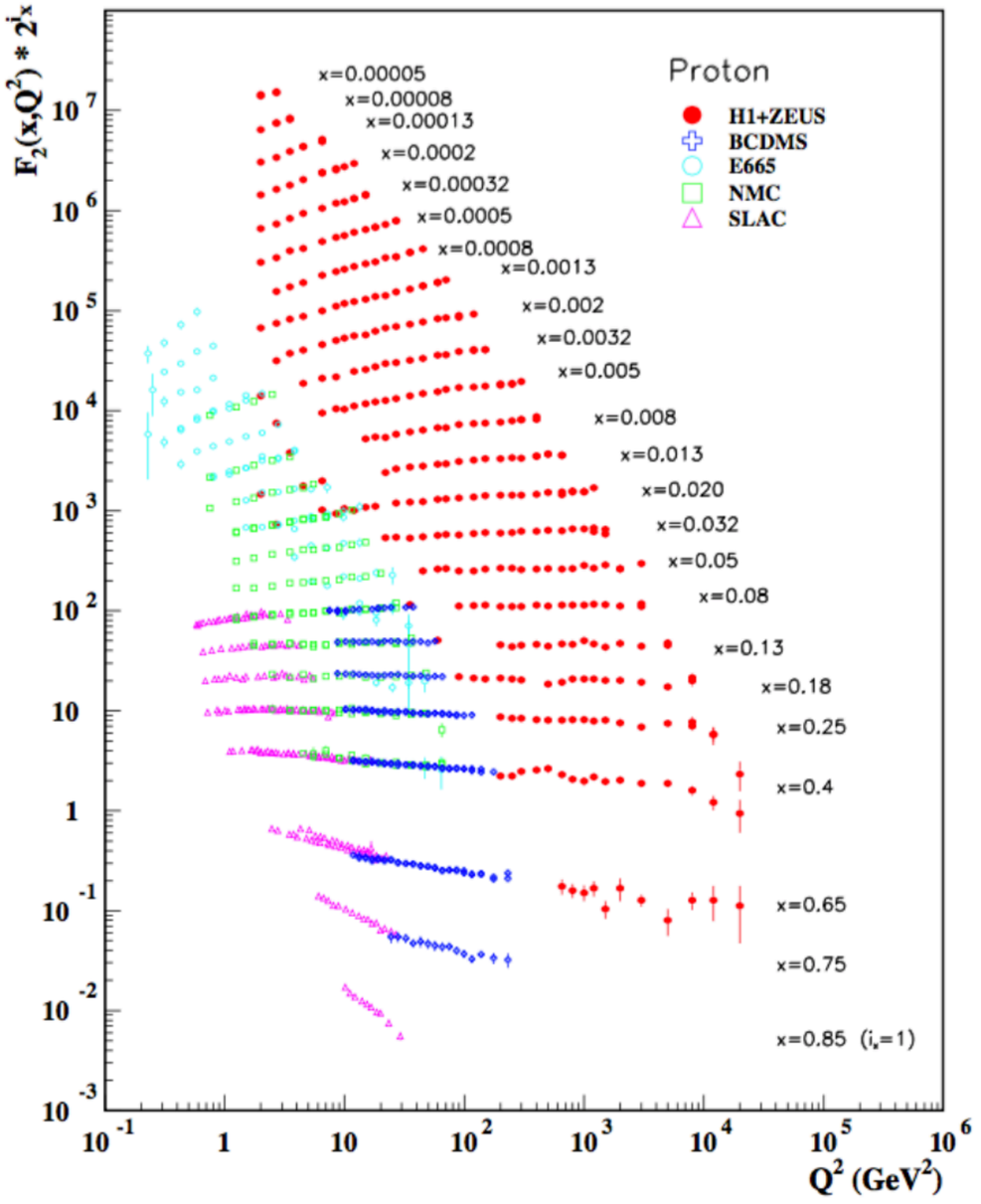


Fig. 9.5: Measurements of the quantity F_2 at increasing values of x as a function of Q^2, compiled by the Particle Data Group (Patrignani *et al.* 2016).

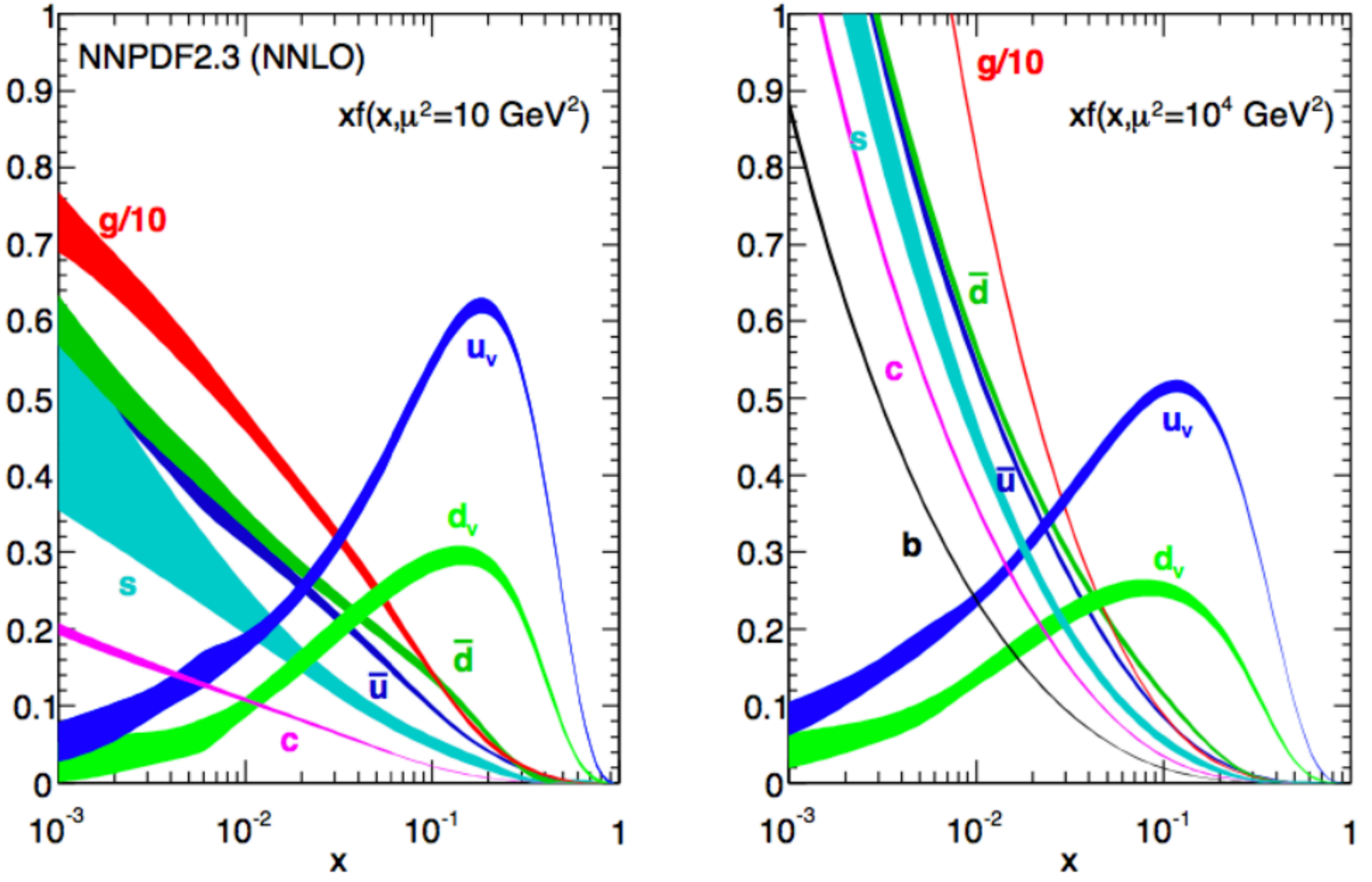


Fig. 10.1: Parton distribution functions $xf_i(x)$ at $Q = 3.1$ GeV and at $Q = 100$ GeV, according to the fit of the NNPDF collaboration, from (Foster *et al.* 2016).

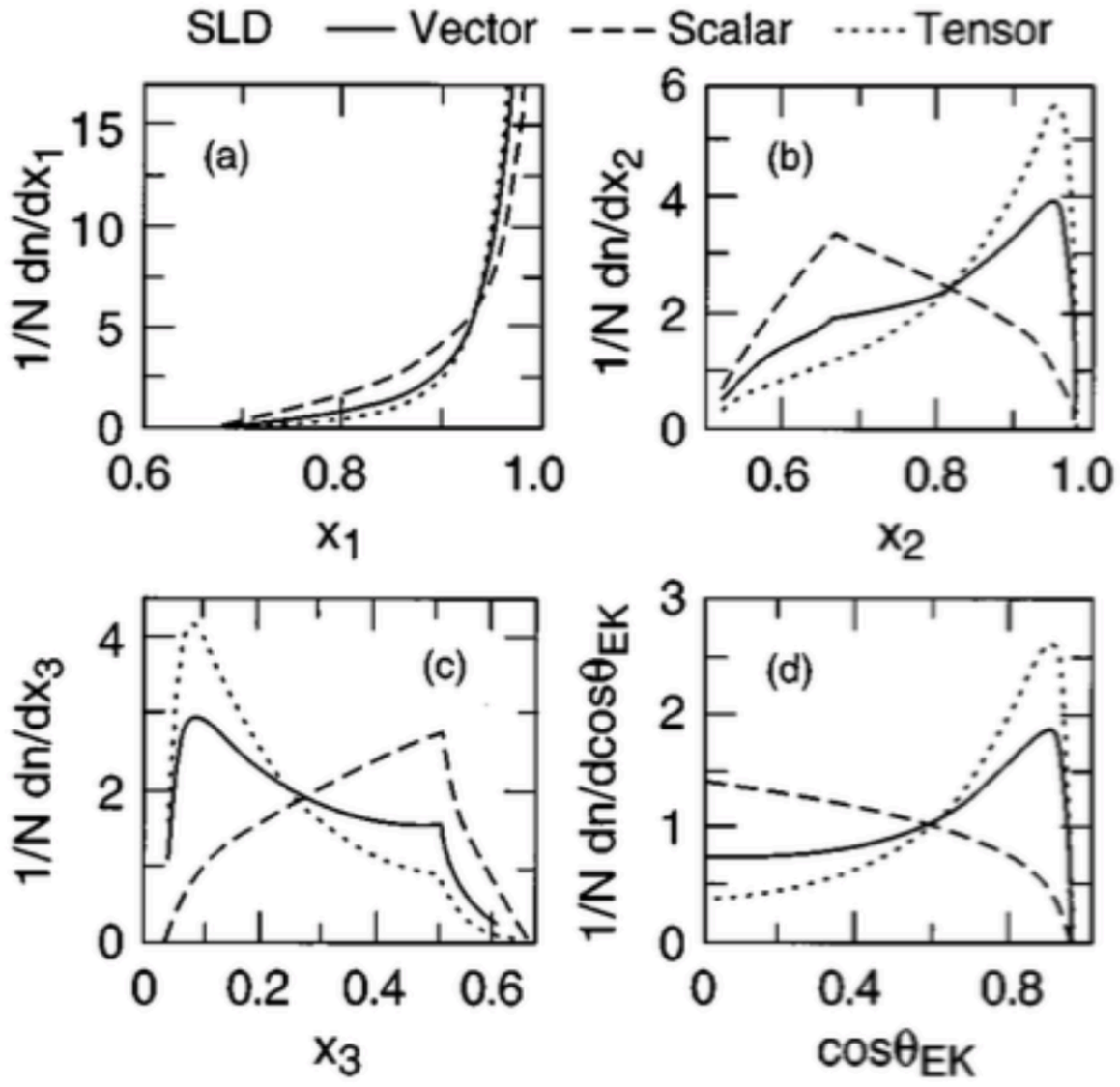


Fig. 10.6: Expectation for the form of the plots in Fig. 10.7 for emission of gluons of spin 0 (scalar), 1 (vector), 2 (tensor), from (Abe *et al.* 1997).

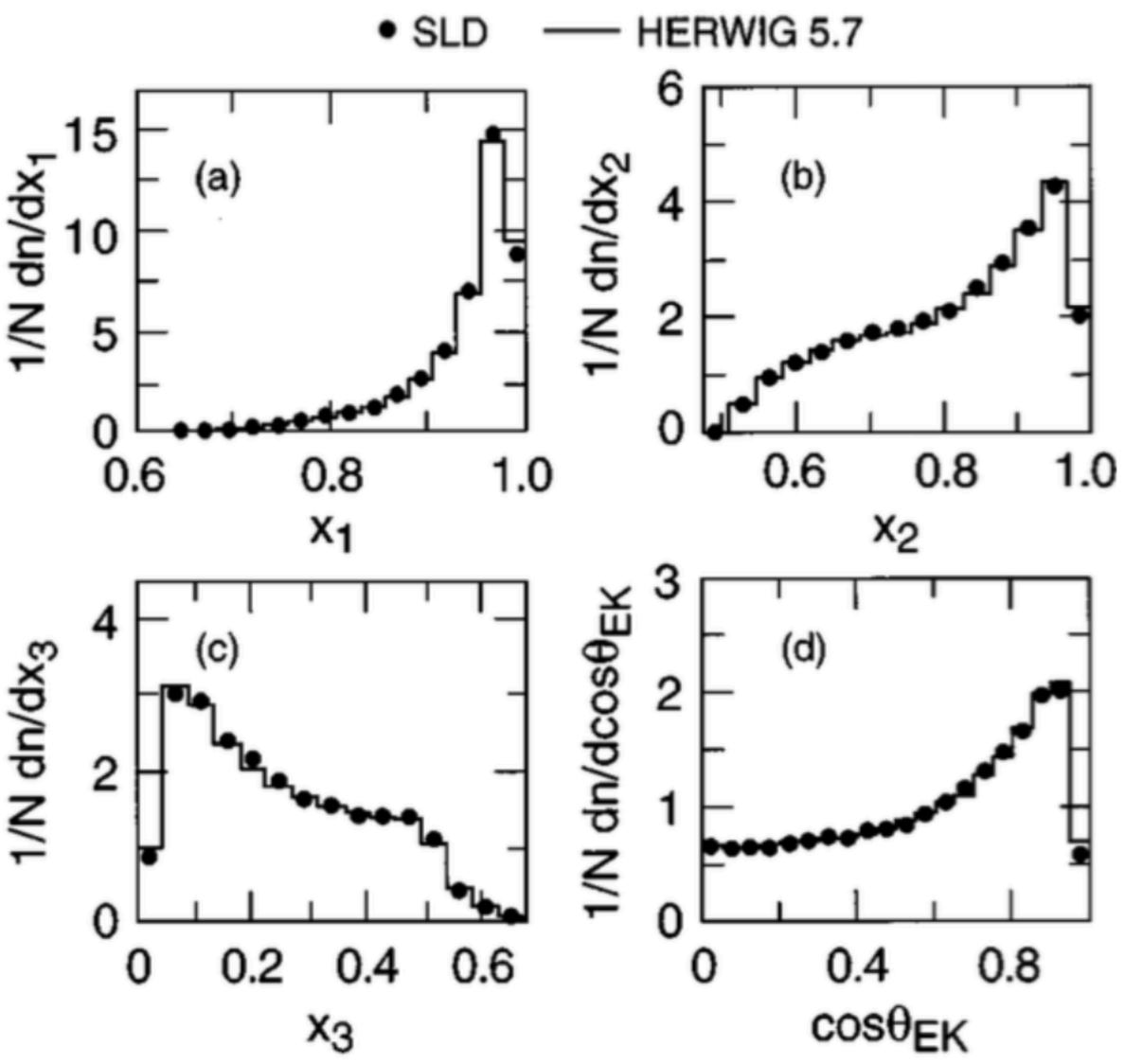


Fig. 10.7: Measurement of the rates of e^+e^- annihilation events as a function of the variables x_1, x_2, x_3 defined in (10.45), (10.54) by the SLD experiment, from (Abe *et al.* 1997).. The data are compared to the simulation program HERWIG, which gives the prediction of the spin 1 gluon model.